JAMES RICHARDSON

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PROFESSIONAL SUMMARY

Experienced Data Engineer with 8 years of hands-on expertise building scalable data pipelines and warehouse solutions. I've spent my career solving complex data problems for fast-growing companies, particularly in fintech and e-commerce sectors. I enjoy mentoring junior engineers and have recently led the redesign of our cloud infrastructure which reduced query times by 60%. Outside of work, I'm passionate about exploring new data technologies and occasionally contribute to internal knowledge sharing sessions.

WORK EXPERIENCE

Senior Data Engineer

DataFlow Systems

dataflow.io

Cambridge, United Kingdom

November 2021 - Present

Designed and implemented Apache Spark-based ETL pipelines processing 2 terabytes daily for 150+ enterprise clients across multiple geographic regions.

Architected a real-time data warehouse migration from Redshift to Snowflake, improving query performance by 58% and reducing infrastructure costs by 42%.

Established data quality frameworks using Great Expectations, reducing production incidents by 75% and improving data governance across teams.

Mentored 4 junior data engineers on best practices in Python development, SQL optimization, and cloud-native architecture patterns.

Data Engineer

Quantum Analytics Ltd

quantumanalytics.co

London, United Kingdom

January 2021 - October 2021

Built and maintained Airflow DAGs orchestrating daily batch jobs and real-time streaming data ingestion from various APIs and databases.

Optimized SQL queries and database indexing strategies, reducing average query execution time from 45 minutes to 8 minutes for critical reporting dashboards.

Developed Python-based data validation tools to monitor data quality metrics and automatically trigger alerts when anomalies were detected.

Collaborated with data science team to build feature engineering pipelines supporting machine learning model training and deployment.

Data Engineer

FinTech Innovations plc

www.fintechinnovations.com

London, United Kingdom

March 2019 - March 2021

Created robust ETL processes using Python and SQL to ingest financial transaction data from multiple payment gateways and legacy systems into data warehouse.

Implemented comprehensive logging and monitoring solutions using ELK stack to track pipeline performance and troubleshoot issues in production environments.

Automated infrastructure provisioning and CI/CD pipelines using Terraform and Jenkins, reducing deployment time by 50% and enabling faster feature releases.

Conducted code reviews and established coding standards for the data engineering team, improving code quality and maintainability across projects.

EDUCATION

BSc Computer Science

University of Manchester

Manchester, United Kingdom

Graduated: June 2016

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, Bash

Big Data Technologies: Apache Spark, Hadoop, Apache Kafka

Cloud Platforms: AWS (S3, EC2, EMR, Redshift), Google Cloud Platform (BigQuery, Dataflow), Azure Data Lake

Data Warehousing: Snowflake, Redshift, PostgreSQL, MySQL

Orchestration and Scheduling: Apache Airflow, Luigi

Data Quality and Testing: Great Expectations, pytest, dbt

DevOps and Infrastructure: Terraform, Docker, Jenkins, Git

Messaging and Streaming: Apache Kafka, RabbitMQ, Kinesis

Monitoring and Logging: ELK Stack, Datadog, CloudWatch